

SIMATS ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED: CO5: Apply computer vision techniques to real-world applications such as face recognition, tracking, medical imaging, and autonomous systems. (BL3)

ASSESSMENT TOOL 3 | Weightage: 20% | Total Marks: 20

Annexure A - Mock Interview

Code of Conduct:

I **Gadwal Mohammad Muzammil - 192424279** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **G. M. Muzammil**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Technical Knowledge	25%	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge	
Problem Solving	25%	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts	
Analytical Thinking	15%	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated	
Communication	15%	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively	
Confidence	15%	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism	
Response to Questions	10%	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately	

1. What is Eigenfaces?

Eigenfaces is a foundational appearance-based 2D face recognition method introduced by Matthew Turk and Alex Pentland (1991). It applies Principal Component Analysis (PCA) directly to vectorized training face images to discover the orthogonal basis vectors (termed 'eigenfaces') that account for the maximum variance across the dataset. Any novel face is recognized by subtracting the mean face, projecting the centered image vector onto the low-dimensional eigenface subspace, and performing nearest-neighbor classification using Euclidean or Mahalanobis distance. It provides substantial dimensionality reduction (e.g., from 4,096 pixels to 50 coefficients).

2. Define PCA in image processing.

Principal Component Analysis (PCA) is an unsupervised linear mathematical technique used for feature extraction, noise reduction, and dimensionality reduction. Given high-dimensional image vectors x_i in R^D , PCA computes the sample covariance matrix $\Sigma = (1/M) \sum (x_i - \bar{x})(x_i - \bar{x})^T$ and solves the eigenvalue problem $\Sigma * v_k = \lambda_k * v_k$. The resulting eigenvectors v_k define an orthogonal coordinate system where components are mutually uncorrelated and ordered by descending variance λ_k . It allows high-dimensional image data to be compressed into a compact subspace while retaining >95% of total signal energy.

3. What is contour-based shape representation?

Contour-based shape representation describes an object purely by its 1D outer boundary perimeter. Prominent techniques include Freeman Chain Codes (encoding directional edge transitions), Polygonal Approximations (Ramer-Douglas-Peucker algorithm), B-Splines, and Fourier Descriptors. The primary advantage is computational efficiency: an object boundary is encoded using $O(K)$ perimeter points instead of $O(M*N)$ area pixels. However, it is sensitive to local boundary occlusions, noise perturbations, and non-rigid topological deformations.

4. What is region-based shape representation?

Region-based shape representation characterizes the entire 2D interior spatial mass and geometry of an object. Core methodologies include Hu Invariant Moments (invariant to translation, scale, and rotation), Compactness / Eccentricity metrics, Quadtree decompositions, and the Medial Axis Transform (MAT / Skeletonization). Region-based methods inherently capture topological holes, inner cavities, and structural symmetry, making them significantly more robust to boundary noise and minor perimeter digitizing artifacts than contour-only techniques.

5. How does an active contour (snake) work?

An active contour (snake), introduced by Kass, Witkin, and Terzopoulos, is an energy-minimizing parametric spline $v(s) = (x(s), y(s))$ that deforms iteratively within an image. The evolution is governed by minimizing the energy functional $E_{\text{snake}} = \int [E_{\text{int}}(v(s)) + E_{\text{ext}}(v(s))] ds$. Internal energy $E_{\text{int}} = (1/2)[\alpha |v'(s)|^2 + \beta |v''(s)|^2]$ enforces curve continuity (elasticity α) and smoothness (stiffness β), while external energy $E_{\text{ext}} = -\gamma |\text{Grad}(G_{\text{sigma}} * I)|^2$ attracts the curve toward high-contrast object boundaries. It converges via Euler-Lagrange numerical differential equations.

6. How does a level set method work?

Level Set Methods (Osher & Sethian) represent an evolving boundary $C(t)$ implicitly as the zero-level set of a higher-dimensional scalar function $\Phi(x, y, t) = 0$, where $\Phi > 0$ inside the object and $\Phi < 0$ outside. The surface evolves according to the Hamilton-Jacobi PDE: $(d\Phi / dt) + F |\text{Grad}(\Phi)| = 0$, where F is a speed

function combining curvature regularization and image gradient stopping terms. The major advantage of level sets over parametric snakes is automatic topological handling, enabling boundaries to split into multiple disconnected parts or merge seamlessly without complex re-parameterization.

7. What is the Hough Transform?

The Hough Transform is a robust feature extraction technique designed to detect parametric geometric shapes (straight lines, circles, ellipses) in noisy edge images. It operates by mapping each image edge point (x_i, y_i) into a sinusoidal curve in a discretized parameter accumulator space: $\rho = x_i \cdot \cos(\theta) + y_i \cdot \sin(\theta)$. Accumulator cells that receive intersecting votes from multiple collinear edge points form local maxima (peaks), identifying the physical presence of lines even under severe background noise and partial occlusions.

8. How are Fourier descriptors used in shape analysis?

Fourier Descriptors represent a closed 2D boundary curve as a 1D complex spatial sequence $s(k) = x(k) + j \cdot y(k)$ and compute its Discrete Fourier Transform (DFT): $a(n) = (1/K) \sum s(k) \exp(-j 2\pi n k / K)$. Geometric invariance is achieved analytically: translation invariance by setting DC term $a(0) = 0$; scale invariance by dividing all coefficients by $|a(1)|$; and rotation/start-point invariance by taking absolute magnitudes $|a(n)|$. Low-order harmonics capture global shape symmetry, while high-order harmonics capture localized boundary roughness.

9. How are wavelet descriptors used in shape analysis?

Wavelet Descriptors apply the 1D/2D Discrete Wavelet Transform (DWT) to object boundaries or shapes, decomposing signals into approximation subbands (LL) and multi-scale detail subbands (LH, HL, HH). Unlike Fourier Descriptors which use infinitely extended sinusoidal basis functions (lacking spatial localization), wavelets provide joint time-frequency localization. This allows wavelet descriptors to isolate both the exact spatial location and scale of localized boundary defects (e.g., chipped gear teeth or burrs) with linear $O(K)$ computational complexity.

10. What is a particle filter?

A Particle Filter (Sequential Monte Carlo) is a non-parametric, non-linear, non-Gaussian recursive Bayesian state estimation algorithm. It represents the posterior probability distribution $p(x_t | z_{1:t})$ using a set of N weighted random state particles $\{x_t^{(i)}, w_t^{(i)}\}$. In visual object tracking, particles propagate according to kinematic motion models and are re-weighted based on visual observation likelihoods (color histograms, deep Re-ID embeddings). It excels at maintaining object identity through severe occlusions, non-linear trajectories, and rapid camera motions.

11. How is foreground-background separation done in surveillance systems?

Foreground-background separation isolates moving objects from static scenes. The industry standard is the Adaptive Gaussian Mixture Model (GMM / MoG, Stauffer & Grimson). Each background pixel is modeled as a mixture of $K = 3$ to 5 Gaussian distributions updating dynamically via online Expectation-Maximization. Incoming pixels differing by more than 2.5 standard deviations from all background distributions are classified as moving foreground, refined via morphological opening to remove noise specks and HSV shadow suppression to eliminate cast shadows.

12. How do multi-camera systems track pedestrians?

Multi-camera tracking systems combine single-camera visual tracking (Kalman / Particle filters) with cross-camera Person Re-Identification (Re-ID). Deep metric learning networks (e.g., ResNet-50 with ArcFace / Triplet Loss) extract compact 128-D appearance embedding vectors for each detected person. A central server matches identities across non-overlapping camera viewpoints using cosine similarity, constrained by calibrated multi-camera ground-plane homographies and spatial-temporal transition matrices.

13. What is a Convolutional Neural Network (CNN)?

A Convolutional Neural Network (CNN) is a deep neural network tailored for grid-structured visual data. It comprises convolutional layers with small learnable sliding kernels (e.g., 3x3), non-linear activation functions (ReLU / SiLU), and spatial pooling/striding layers. CNNs enforce two fundamental inductive biases: translation equivariance (patterns are detected regardless of spatial position) and local connectivity (pixels in local neighborhoods are strongly correlated). They learn hierarchical visual representations ranging from low-level edges to high-level semantic object categories.

14. How does a Vision Transformer (ViT) work?

A Vision Transformer (ViT, Dosovitskiy et al.) adapts the standard Transformer architecture for computer vision without using convolutions. An image is divided into non-overlapping spatial patches (e.g., 16x16 pixels), flattened into 1D linear embeddings, prepended with a learnable [CLS] token, and augmented with 1D learnable positional encodings. The patch sequence is processed through stacked Multi-Head Self-Attention (MHSA) and MLP blocks, modeling global all-to-all patch interactions across the entire image at every layer.

15. What is a Generative Adversarial Network (GAN)?

A Generative Adversarial Network (GAN, Goodfellow et al.) is an unsupervised generative framework comprising two competing sub-networks trained via a zero-sum game: a Generator G that maps random latent noise z into synthetic photorealistic images $G(z)$, and a Discriminator D that classifies whether an image is real (from training data) or fake (from G). They optimize the minimax objective $\min_G \max_D E[\log D(x)] + E[\log(1 - D(G(z)))]$, reaching a Nash equilibrium where G produces images indistinguishable from real distributions.

16. How is a Support Vector Machine (SVM) used for image classification?

A Support Vector Machine (SVM) is a supervised learning classifier that finds the optimal separating hyperplane maximizing the geometric margin between different class feature vectors: $\max (2 / \|w\|)$ subject to $y_i(w^T x_i + b) \geq 1$. When image features (HOG, SIFT, color histograms) are not linearly separable in the original input space, the Kernel Trick maps features into a high-dimensional Hilbert space using non-linear kernels (e.g., RBF kernel $K(x, z) = \exp(-\gamma \|x - z\|^2)$) to achieve robust classification.

17. How are Hidden Markov Models (HMM) applied in activity recognition?

An HMM $\lambda = (A, B, \pi)$ models human activity as a doubly stochastic sequential process. The hidden states represent sequential biomechanical motion sub-phases (e.g., Stance $\rightarrow$ Swing $\rightarrow$ Heel-Strike in walking), while emission probability distributions B model continuous observation vectors (joint coordinates, optical flow). The Baum-Welch algorithm trains model parameters, and the Viterbi algorithm classifies unseen video sequences by determining the maximum likelihood hidden state sequence.

18. What is a Gaussian Mixture Model (GMM)?

A Gaussian Mixture Model (GMM) is a parametric probabilistic model representing complex probability density functions as a weighted sum of M Gaussian components: $p(x) = \sum_{m=1}^M c_m * N(x; \mu_m, \Sigma_m)$, where c_m are non-negative mixture weights summing to 1. Model parameters (weights c_m , means μ_m , and covariance matrices Σ_m) are iteratively estimated from data using the Expectation-Maximization (EM) algorithm, widely used for color image segmentation, texture clustering, and background subtraction.

19. How is graph-based learning applied to 3D shape correspondence?

3D surface meshes are non-Euclidean geometric structures modeled as graphs $G = (V, E, W)$. Graph Neural Networks (e.g., MeshCNN, DiffusionNet) utilize spectral graph convolutions $g_{\theta} * x = U * g_{\theta}(\Lambda) * U^T * x$ based on the Laplace-Beltrami operator (LBO). By computing continuous functional maps between spectral eigenbases, graph learning matches anatomical landmarks across non-rigidly deforming 3D shapes (e.g., human faces), invariant to isometric posture changes and mesh resolution variations.

20. How is road and lane detection performed in autonomous driving?

Autonomous road and lane detection employs a multi-stage computer vision pipeline: 1. Inverse Perspective Mapping (IPM) transforms front-camera video into top-down bird's-eye view; 2. Illumination-invariant color thresholding (HLS/LAB space) and Canny edge filtering isolate white and yellow lane markings; 3. RANSAC polynomial curve fitting (or deep semantic segmentation networks like YOLOP/LaneNet) fits 2nd/3rd-order lane polynomials; 4. Temporal Kalman filtering smooths lane trajectory estimates across consecutive frames to guide autonomous vehicle steering.